Image Forgery Detection Using Machine Learning Approach

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***Abstract*— The prominent use of social networking services had led to the ease of manipulating images using various image editing tools, AI tools which has led to the rise of forged images, due to this a threat has been created in the society. As a result of this many cybercrimes have evolved. One of them is deep face masking in which fake images and videos are used. With the help of deep face masking technique one can superimposes one's identity. The combination of image forgery and deep face masking could make the risks that are associated with cybercrime and privacy violations. There are many deep learning (DL) models which have been trained using convolutional neural network architecture for detecting forged images. The proposed system uses analysis of Machine Learning (ML) models with the help of error level analysis. This paper mainly describes about various image forgery methods and later it shows the comparison of different machine learning algorithms out of which Random Forest performed well with an accuracy of 93.25%.**

Keywords— Image forgery, Machine Learning, Deep learning, deep face masking, random forest.

I. Introduction

The tremendous use of internet and social media are the reason why forged images have been widely evolved in many aspects of social media and use of the in many areas like news, advertisements and many more. Many AI tools and photo editing tools also led to the development of tampered images. As a result of this new challenges in the form of cybercrimes, including image forgery and manipulation has evolved. One such example for it is deep face masking [1]. Deep face masking involves the development of forged images which manipulate the individuals in a form which helps them gain some money from the individuals.

Due to these types of forgery, it may lead to the spreading of misinformation and reputation damage to the popular people on social media. As the technology is being developed it is making easier people to make various attacks and passing misinformation to people through social media [2]. The whole world revolves around the images. Every object that can be perceived by the human eye is represented as a visual image. But the use of digital images introduced the threat of forged images. It is challenging for individuals to recognize any alterations made to an image. These type of image forgeries results in vast implications on internet [3]. Today the world is revolving on the internet and lots of digital data is being generated due to it. Image is also a part of digital data and is the main reason behind threats like deep face masking [4]. Due to all these crises new methods are getting introduced to efficiently detect the manipulated images from the original one. The precision of detecting the forged images depend on how effectively one train the model with help of wide range of data.

There are many image forgery detection techniques. Some of the methods are passive and some of the are active methods. Here are the examples of passive and active methods include copy-move detection, error level analysis (ELA), noise analysis, watermark analysis, deep fake detection etc. Copy-move forgery is an image manipulation technique that involves the duplication and insertion of one or more regions from the same image into different locations [5]. It is the most commonly used technique to manipulate the images. It can be easily done with the help of different editing tools. This technique works by analyzing patterns and features in the image to analyze duplicate content. Copy-move forgery make significant challenges in the image forensics [6]. As many AI tools are being introduced, these are making the task easier to manipulate the images.

Error level analysis method involves identifying areas within an image that are at various compression levels. ELA is based on lossy image compression. Especially jpeg formats can be applied well using this method [7]. The use of error level analysis and then rather apply machine learning algorithms on them so that the accuracy becomes precise. ELA mainly helps in image compression. When an image is compressed, it introduces different noise patterns in it. These noises differ from image to image and also regions to regions in it. While ELA helps in the detection of noise in a image on the other hand machine learning can classify the output of ELA into forged images or not. The combined use of machine learning with error level analysis can help to improve accuracy of the algorithm and can adapt the model to train on diverse datasets. Use of diverse datasets can help to make more accurate predictions.

II. Related Work

Digital images are becoming the reason in rising the threat of image manipulations. The field of machine learning and deep learning are the emerging technologies which are best suited for image classification and image detection [8]. Yet new trends and technologies are being developed the threat of image manipulation is also increasing very fast.

Ying Zhang et al. [9] has worked on Image Region Forgery Detection using deep learning. He used CASIA dataset and obtained accuracy of 91.09%. He included Autoencoders in his network to detect the tampered regions. Survey on Image Forgery Detection using deep learning concludes that deep learning methods reported better accuracy than the traditional algorithms [9]. Work deployed using CNN recompressed

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images have gained a validation accuracy of 92.23% [10]. Agarwal et al. [11] has worked on deep learning techniques that are employed on copy-move forgery. He mainly focuses on feature extraction using deep learning and then using of copy move detection technique for classifying images. The algorithm used by them initially takes an input image and detect the forged regions in it. It includes feature extraction and preprocessing of the image and at last identifying the forged regions. A. B. Z. Abidin et al. [12] has studied on different deep learning approaches in the detection of copy-move image forgery.

Sadeghi et al. [13] aimed to evaluate various detection methods for copy-move forgery considering factors like jpeg compression, scaling, and rotation, also discussed the significance of each technique in the current image forensics. Chen et al. [14] introduced a method for digital image forgery detection. The architecture used by them includes 23 layers and they presented a filter layer as the input layer which has RGB colour components and direction information as block features [15]. The function works together with trigonometric transformations to detect the manipulated parts of the image. It also involved CNN for forgery detection [16]. Radom Walker segmentation method for locating tampered regions and then used Long short-term memory (LSTM) based network architecture for classification and localization. Work involving two key aspects that are the use of convolutional neural networks and the evaluation of transfer learning with pre-trained ImageNet Models using the CASIA v2.0 dataset is widely adopted in recent past. The most popular approach of detection of copy-move forgery using deep learning methods uses pre-trained ImageNet model and then done fine-tuning its structure with small training samples. P. Yang et al. [16] mentioned that due to the increasing speed of internet it has impacted in growing of forged images. He combined the deep learning along with Error Level Analysis (ELA) technique for document image forgery. It not only detects the document forgery but also shows the forged text areas. A. Kuznetsov et al. [17] addresses the image forgery in remote sensing data for safety. He used convolutional neural network architecture and it gave an f1 metric of 0.77. He also mentioned that his approach can be used in splicing detection. A. Korkmaz et al. [17] highlights the increasing fake images through digital media. He employed three parallel deep neural network structures and three various datasets used in it. V. Sharma and N. Singh worked on copy-move forgery. They presented a deep convolutional neural network than combines a 50-layered residual network. The residual network uses skip connections and handles the issues of vanishing and exploding gradients in deep learning models [18].

III. Methodology

There are several pre-existing methods employed for image forgery detection. Some of the techniques are active and some of them are passive. Active image forgery detection includes: Digital Watermarking, Digital signature. Passive image forgery detection techniques include: Pixel-based technique, error level analysis, copy-move detection, source camera identification. Fig.1 presents the proposed method.

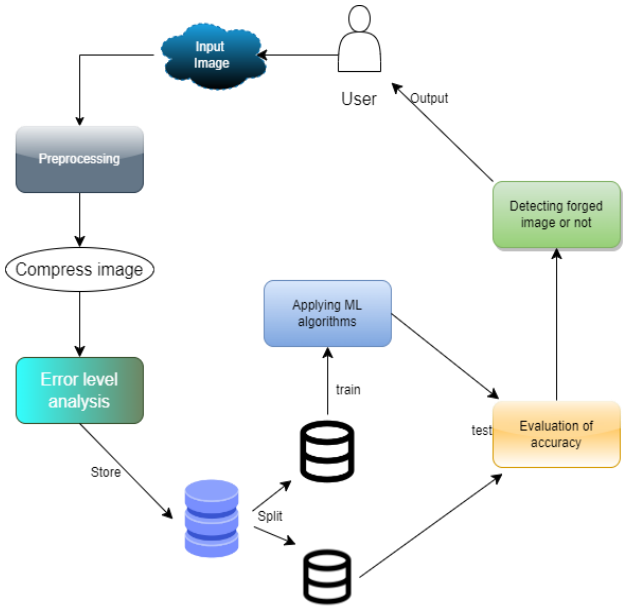

Fig. 1. Architecture of the proposed system.

The process of identifying whether an image is authentic or manipulated is described in the diagram above. The procedure begins with the user inputting an image, which then undergoes pre-processing. Pre-processing involves loading the image and preparing it for feature extraction. Feature extraction is the process of gathering relevant information from the image using various techniques, such as Error Level Analysis. This method examines the image and identifies any discrepancies that may have arisen due to digital manipulation. By following these steps, it is possible to determine whether the image is genuine or has been edited in some way.

Further ML algorithms are used for train the new image dataset. Some algorithms that can be used are decision tree, random forest, adaboost, naïve bayes, support vector machines etc. Then, the algorithm then prepared to validate the test data. After this algorithm with highest accuracy is chosen to test for new sample of data. Using the training experience, the model tells whether the given image is forged or not.

The advantage of Error Level Analysis and Machine Learning which results in precise accuracy. Error Level analysis does a sequence of steps that include: compression analysis, error level calculation, visualization, and analysis. With the help of error level analysis one can identify whether any changes happened in an image or not. The combined effect of error level analysis and machine learning can enhance the accurate prediction of the forged images. Use of both error level analysis and machine learning make the ability to identify the subtle replacements that might be difficult to identified by other techniques.

A. Dataset Description:

The dataset is taken from Kaggle, where it consists of 2000 images. Out of which 75% i.e. 1500 images are used for training and the rest of the images are used to test the model.

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B. Algorithms analyzed:

Decision Tree: It serves as a supervised learning model that is applicable to both classification and regression tasks. Most of the times it focuses on classification tasks. It simply has a tree like structure, it consists of a root node, internal nodes and leaf nodes. It used CART algorithm to construct a tree structure. Entropy and gini are the criterion used in Decision Tree. The default value is "gini", but also "entropy" can be used.

SVM: Support Vector Machines is one of the mostly used Supervised Learning model applicable for both Classification and Regression problems. SVM creates a hyperplane by choosing the points that behave in more similar manner. Those points that are used to create a hyperplane are called as support vectors.

Random Forest: This algorithm belongs to the supervised learning technique, which addresses the tasks of classification and regression. It builds multiple decision trees, and the majority one is chosen for classification. The algorithm is based on ensemble learning and is a bagging algorithm.

Adaptive Boosting: It is a popular ensemble learning algorithm used for both classification and regression. It combines the weak learners usually decision trees to create a strong learner. It assigns weights to training instances which gives more importance to misclassification samples. This algorithm mainly aims to correct the errors of previously used algorithms and make it robust and precise.

Naive Bayes: It is a supervised learning model based on Bayes Theorem. It assumes there is an independence among variables. It predicts the class of the unknown variables of dataset based on bayes theorem. It is mainly used for large datasets.

$$P(c|x) = \frac{P(x|c)*P(c)}{P(x)} \quad (1)$$

$$\mathrm{P(c|x)} = \text{Posterior Probability} \quad (2)$$

$$\mathrm{P(x|c)} = \text{Likelihood Probability} \quad (3)$$

$$\mathrm{P(c)} = \text{Class Prior Probability} \quad (4)$$

$$\mathrm{P(x)} = \text{Predictor Prior Probability} \quad (5)$$

$$\mathrm{P(c\,|X) = P(x1|c) * P(x2\mid c) * \ldots * P(xn\mid c) * P(c)} \quad (6)$$

where: c is the occurrence of event c, x is the occurrence of event x

The proposed algorithm is given below.

Algorithm: Image forgery detection using error level analysis and machine learning

Input: Set of N images(I), testing images(test)
Output: Prediction for forged images(y_pred)
Training Phase
for each x in I **do**
    Apply error level analysis to x
    Extract features from error level analysis result
    Label x as forged image or not
    Store the image in a training dataset
end for
Train a machine learning model using training dataset
Classification Phase
Initialize: y = []
for each y_i in test **do**
    Apply error level analysis to y_i
    Predict y_i using the model that is used for training
    Append y_i to y
end for
Output y

IV. Results And Discussion

Some of the metrics that are used to test how an algorithm is efficient or not include:

Confusion matrix: It is a square matrix of size N, where N is the number of predicted samples. It is a combination of actual and predicted values. It mainly focuses to calculate precision, recall, accuracy and f1score.

Accuracy: It represents the number of correctly classified samples of a data to the overall samples of a data.

$$\text{Accuracy} = \frac{\mathrm{TP+TN}}{\mathrm{TP+FP+TN+FN}} \quad (7)$$

Precision: It becomes 1 when TP = TP+FP, it ideally implies FP = 0. If FP value increases the overall value automatically gets decreased.

$$\text{Precision} = \frac{\mathrm{TP}}{\mathrm{TP+FP}} \quad (8)$$

Recall: It is similar to precision but it calculates the false negative sample rate for a model.

$$\text{Recall} = \frac{\mathrm{TP}}{\mathrm{TP+FN}} \quad (9)$$

F1 Score: The calculation denotes the harmonic mean of precision and recall. This considers both false positives and false negatives which makes it effective for performing well on imbalanced datasets.

$$\text{F1 Score} = \frac{2*\mathrm{Precision}*\mathrm{Recall}}{\mathrm{Precision+Recall}} \quad (10)$$

Using the metrics that are mentioned in equations 7,8,9 and 10 and with the help of the dataset that is described in the section III, the following are the results obtained from it. The below table describes the comparison of different metrics that are evaluated using machine learning algorithms. Table 1

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presents the comparison of several algorithms with proposed method using several metrics.

TABLE I. EVALUATION MATRIX

Algorithm	Accuracy	Precision	Recall	F1 Score
SVM	90.05%	96.23%	83.58%	89.7%
Decision Tree	82.25%	85.32%	78.1%	82.03%
Random Forest	93.25%	93.93%	92.53%	93.22%
Naïve Bayes	84.54%	76.86%	97.51%	85.96%
AdaBoost	82.55%	87.42%	76.11%	81.37%
Proposed method	**94.62%**	**95.98%**	**97.97%**	**96.96%**

It could be clearly observed that random forest performed well with an accuracy of 93.25% and Decision tree performed with a least accuracy of 82.25%. The proposed method uses the advantage of error level analysis technique along with random forest which as a result gave an accuracy of 94.62%. The error level gave an upper hand to random forest to improve its accuracy because, it is built on multiple decision trees.

Fig. 2 presents the pie chart depicting the time taken by each algorithm in order to train the model using the dataset. It is clearly observed from above pie chart that Naïve bayes took less time for training. As, the model knew for its fast training and relies on bayes theorem for computation it helps the model for fast training. Although random forest gave a good accuracy because of it's multiple decision trees it took more time when compared to Naïve bayes. Naïve bayes make independent assumptions between the features. As a result of it indirectly reduces the complex computations that need to be done. Fig. 3 presents the sample image (original and forged).

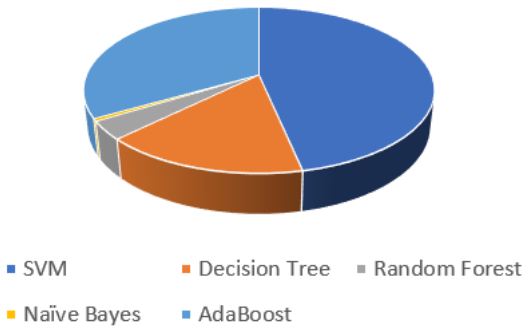

Fig. 2 Time taken by the algorithms to train the model.

Fig. 3 Left to right (Original and forged image).

V. CONCLUSION

It can be concluded that in-order to detect the forged images for large datasets efficiently, the proposed model that is random forest can be used along with error level analysis technique for image forgery detection which resulted in a decent accuracy. This paper compared the results of models and among these random forest gave a good performance. All the algorithms also performed good but, random forest gave a good accuracy score compared to all other algorithms. It is noted that random forest learning algorithm combines multiple decision trees, this makes it less prone to the chance of overfitting and can also deal with complex data. The pre-existing technique error level analysis helped in identifying and reduce the misclassification and errors which makes the model more accurate and robust.

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